

ABSTRACT

WEB-BASED LARGE PRIME VALIDATION SERVICE

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ABSTRACT

The **Web-Based Large Prime Validation Service** is a web application developed to efficiently determine whether a given large integer is prime or composite. Prime number validation is an important concept in number theory, algorithm design, and cryptography. Although small numbers can be checked easily using basic methods, validating large integers through traditional trial division can require a considerable number of computations. The proposed system addresses this challenge by providing a simple web-based interface combined with an efficient primality-testing approach.

The application allows users to enter a large integer through a browser interface and receive a clear validation result. Before the number is processed, the system performs basic input validation to ensure that the entered value is numeric, positive, and within the supported range. After validation, the number is passed to the primality-testing module. The system uses modular arithmetic and modular exponentiation to perform the required calculations more efficiently than checking every possible divisor individually.

The main algorithm considered in the project is the **Miller–Rabin primality test**. It is a fast primality-testing method based on modular arithmetic. The algorithm represents $n - 1$ in the form $d \times 2^s$, where d is odd, and then performs modular exponentiation using selected bases. The results are examined according to the conditions of the test. If a number fails the required conditions, it can be identified as composite; numbers that pass the selected tests are treated as probable prime within the scope of the implementation.

The web-based nature of the service makes the primality-testing process accessible through a straightforward user interface. The frontend can be implemented using standard web technologies such as **HTML, CSS, and JavaScript**, while the backend performs the core large-integer processing. The separation between the user interface and the computational logic allows the system to be organized clearly and provides a foundation for further improvements. The service can therefore demonstrate both algorithmic concepts and their practical use in a software application.

One of the main advantages of the proposed approach is improved computational efficiency when compared with straightforward trial division for large values. Instead of attempting to test every possible divisor, the primality-testing process uses repeated modular exponentiation to reduce the amount of unnecessary computation. This makes the approach more suitable for a web application where users expect a result without a long waiting period. The project also demonstrates how choosing an appropriate algorithm can directly affect the performance of an application.

The system can be useful as an educational platform for understanding prime numbers, modular arithmetic, primality testing, and algorithm efficiency. It also provides a basic foundation for applications related to cryptography, where prime numbers play an important role. The service can be extended in the future with improved large-integer handling, optimized computation, stronger input validation, caching, asynchronous processing, and an API that allows other applications to use the validation service.

Overall, the **Web-Based Large Prime Validation Service** provides a practical and user-friendly method for validating large integers. By combining a web interface with the Miller–Rabin primality-testing approach, the project demonstrates the practical application of number theory and efficient algorithm design. It provides a useful base for further development into a more advanced large-number validation or cryptography-oriented service.